

Model Selection

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1. Why Use a Validation Set?

- The **training set** is used to learn the model parameters.
- The **validation set** helps us evaluate how well the model generalizes to unseen data.
- If we choose the model based solely on training loss, we risk picking one that has memorized the data instead of learning general patterns.

2. Model Selection Process Using a Validation Set

At each iteration of gradient descent:

1. **Train the model** on the training set.
2. **Compute training loss** $J_{\text{train}}(\theta)$.
3. **Evaluate the model** on the validation set and compute **validation loss** $J_{\text{val}}(\theta)$.
4. **Monitor validation loss:**
 - If validation loss decreases, the model is improving.
 - If validation loss starts increasing, **overfitting has begun**, and we should stop training.

3. Choosing the Best Model

- Instead of selecting the last model from gradient descent, we choose the model that has the **lowest validation loss** before overfitting starts.
- This can be done by keeping track of the validation loss at each step and saving the model parameters that achieved the **best validation performance**.